

# Above-Ground Biomass Estimation

A Case Study in Lopé, Gabon

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# Why it is necessary to estimate biomass?



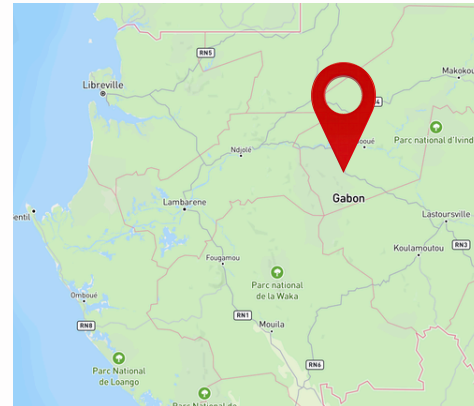
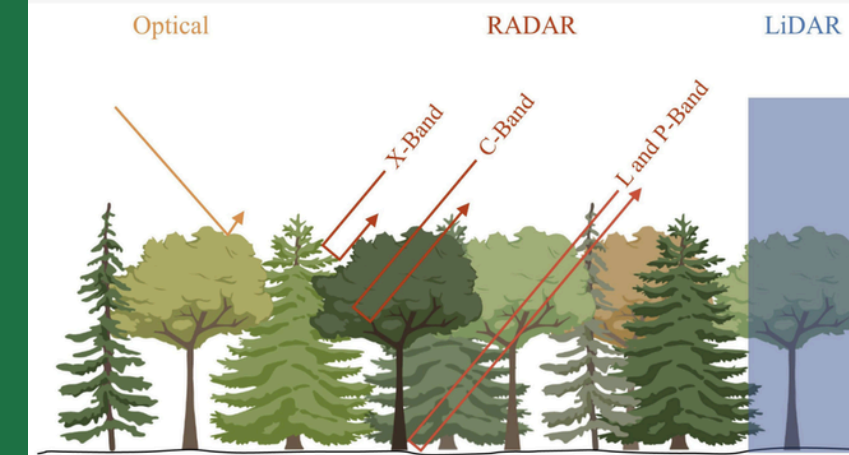
Tropical forests provide 50% of the total plant biomass of the Earth, and are renowned as 'the lungs of the planet Earth' due to their role in the exchange of gasses-particularly inhaling CO<sub>2</sub> and breathing out O<sub>2</sub> within the atmosphere (*Duncanson et al., 2020*)

AGB is a vital indicator of forest condition, directly associated with carbon dynamics, ecosystem productivity, and biodiversity (*Ma et al., 2024*)



# STUDY AREA

# DATASET USED



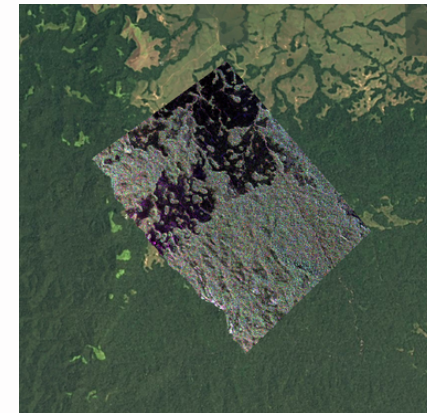
Location: Lopé National Park, Gabon



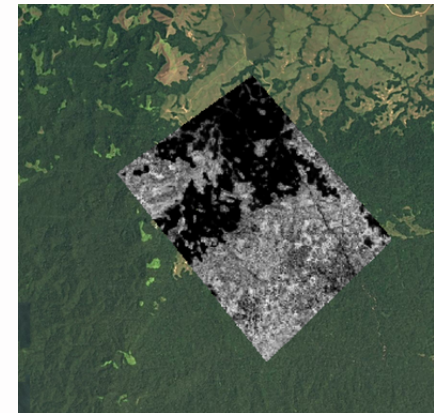
Ecosystem: Tropical rainforest



FSAR L Band,  $\gamma^0$  backscatter (2 m)



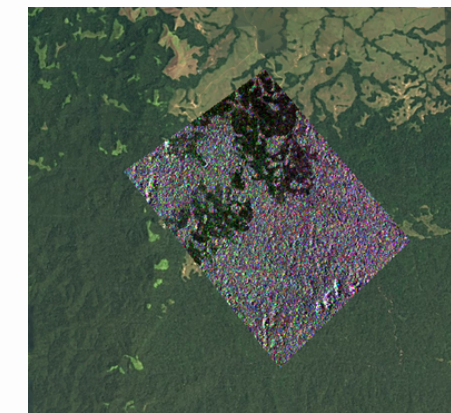
FSAR P Band,  $\gamma^0$  backscatter (2 m)



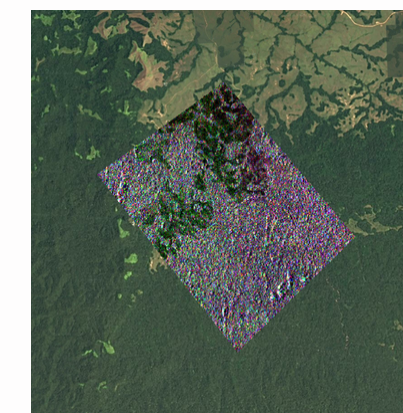
LVIS LiDAR-derived AGB, 50m



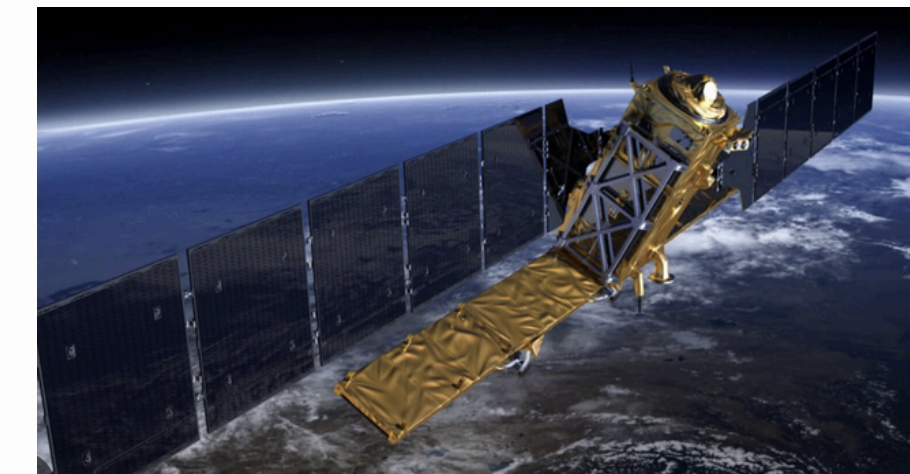
Airborne (FSAR & LVIS), EPSG: 32732



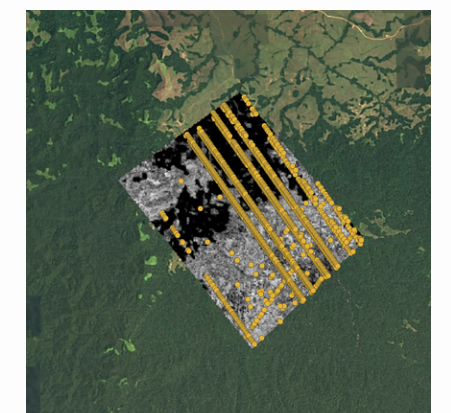
Sentinel VV



Sentinel VH



Sentinel-1, C Band, 30 m, 15n,  $\gamma$ , EPSG: 32732



GEDI LiDAR footprints



GEDI, 25 m, EPSG: 4326



# OBJECTIVES

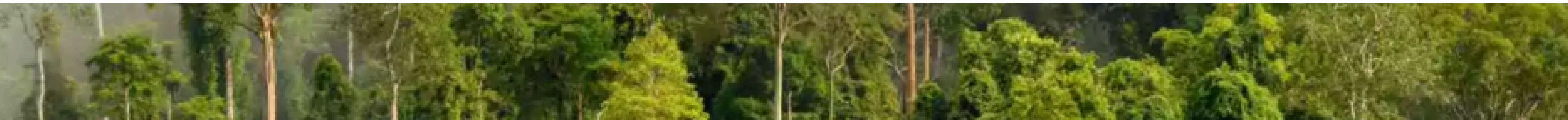
Estimate forest aboveground biomass (AGB) using multi-frequency SAR observations (C-, L-, and P-band) with GEDI biomass as reference

Quantify the relationship between SAR-derived metrics (backscatter  $\gamma^0$ , cross-ratio, RVI) and GEDI footprint-level AGB

Compare the performance of linear regression and random forest regression for AGB estimation using cross-validation

# GOAL

Evaluate the capability of multi-frequency SAR observations to estimate forest aboveground biomass



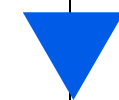


# WORKFLOW



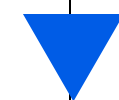
## PRE-PROCESSING

Multi-temporal aggregation (median composite) of SAR backscatter  
SAR Metrics: Derivation of Cross-Ratio and Radar Vegetation Index (RVI)  
Reprojection of GEDI footprints to match SAR coordinate system  
Resampling of L- and P-band SAR data to common spatial resolution  
Creation of predictor variables for modelling



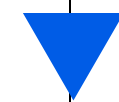
## CORRELATION ANALYSIS

Extraction of SAR features at GEDI footprint locations  
Spearman rank correlation between SAR metrics and GEDI AGB



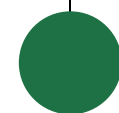
## MACHINE LEARNING

Linear regression (baseline) with 5-fold cross-validation (single predictors per band)  
Random Forest regression with 5-fold cross-validation (multi-predictor, L+P bands + indices)



## AGB ESTIMATION

Wall-to-wall AGB mapping using the best-performing model

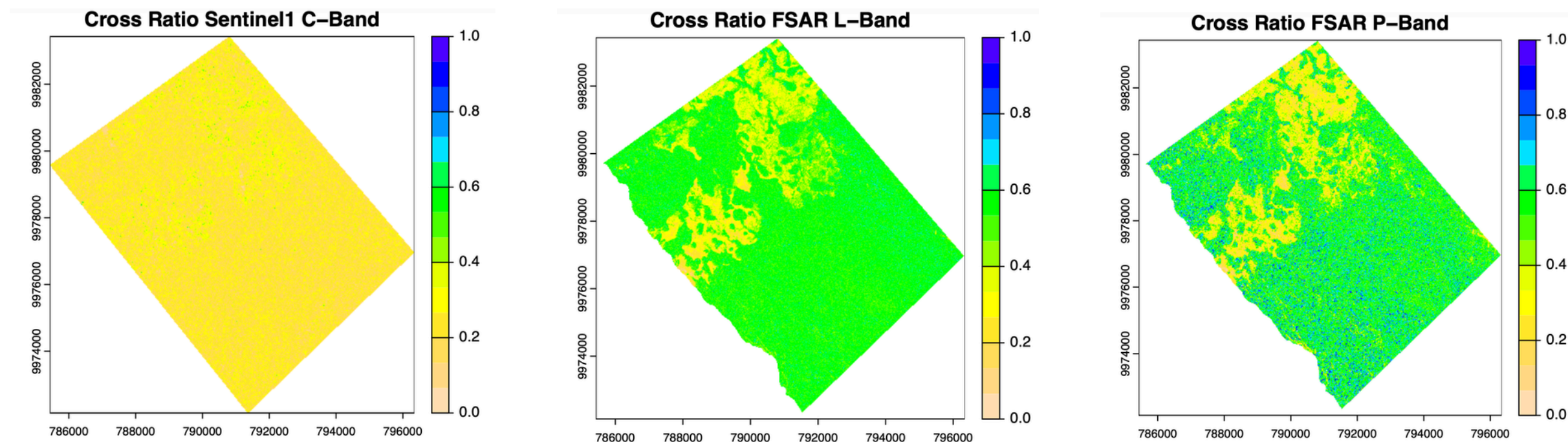


## VALIDATION

Pixel-wise comparison against LVIS AGB reference map  
Performance metrics:  $R^2$ , RMSE, MAE

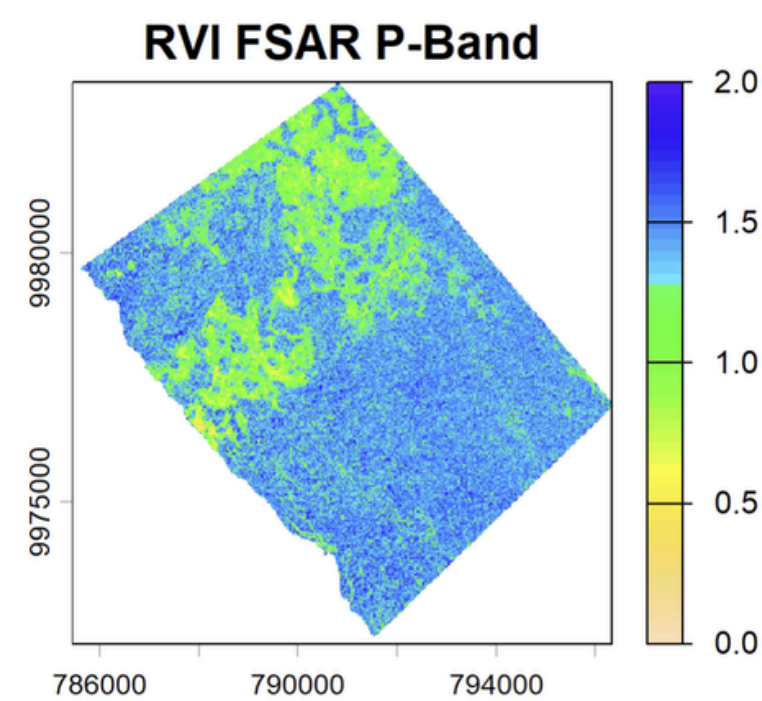
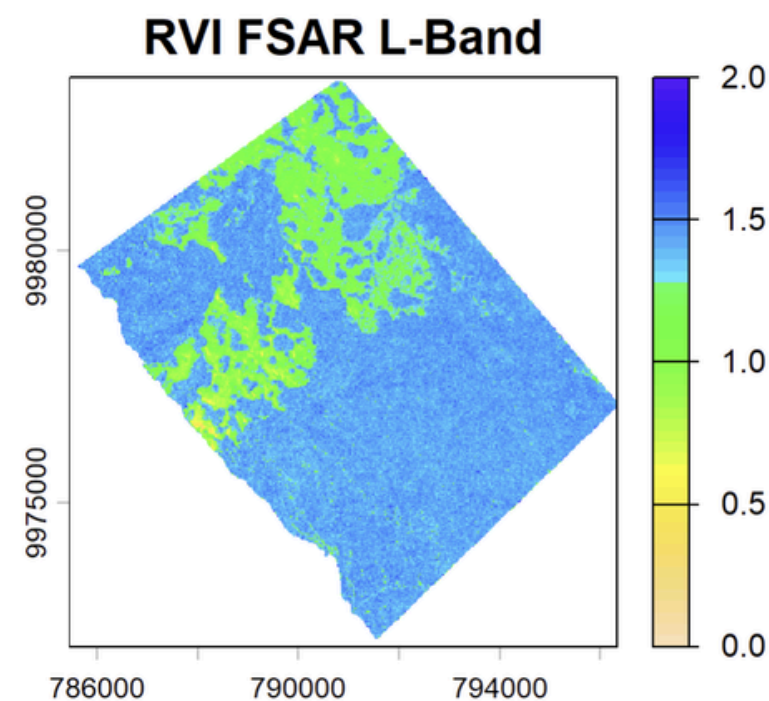
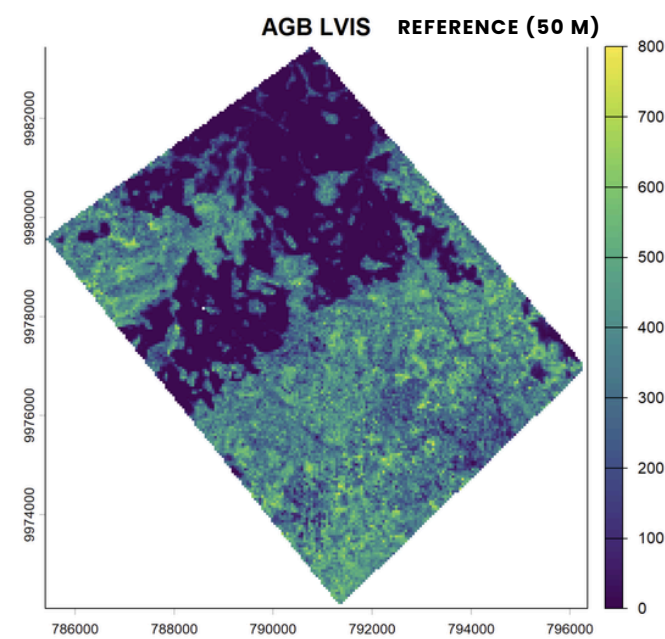


# SAR VEGETATION INDICES for AGB ESTIMATION



Lower AGB sensitivity

Higher AGB sensitivity



## INTERPRETATION

**Low values** ( $CR < 0.2$ ,  $RVI < 0.3$ ): sparse vegetation or open areas.

**Moderate values:** shrubs, young forest, low biomass

**High values** ( $CR > 0.5$ ,  $RVI > 1.0$ ): mature forest, high biomass

## KEY RESULT

L- and P-band indices show higher sensitivity to assess AGB than C-band due to deeper canopy penetration

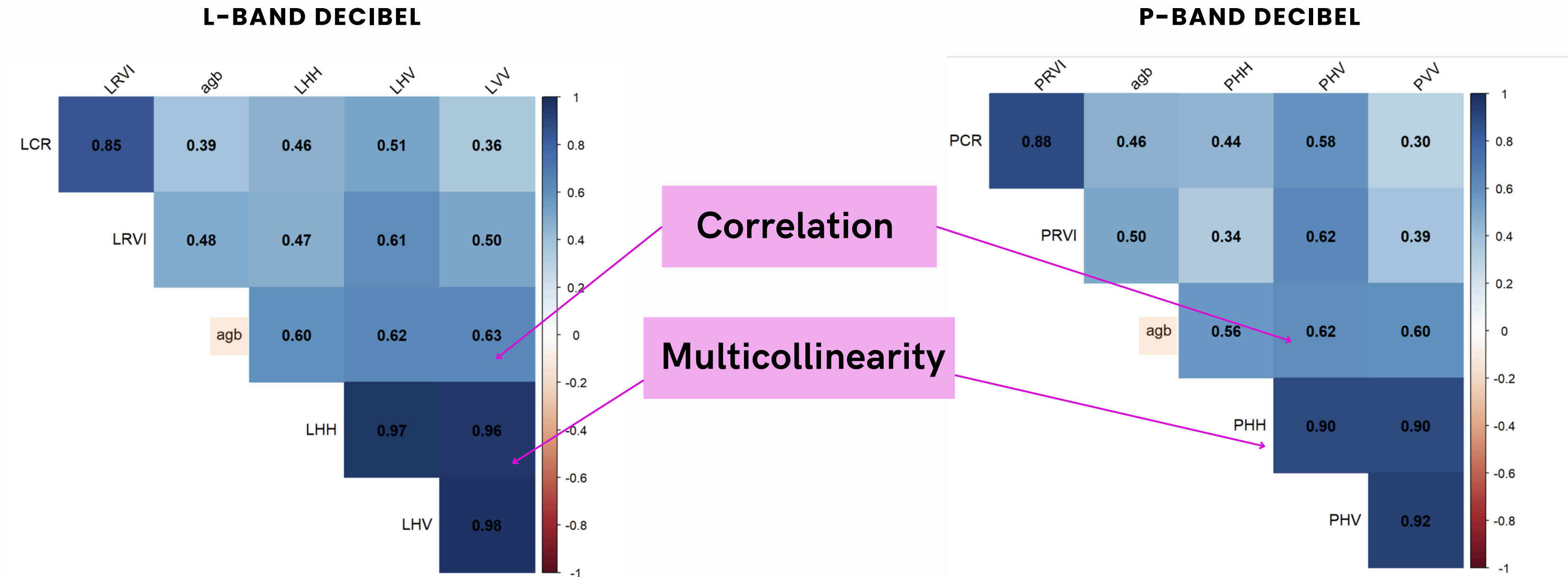
$$CR = VH/VV$$

Indicates volume scattering intensity

$$RVI = 8 \cdot HV / (HH + VV + 2 \cdot HV)$$

Estimation of the randomness of scattering

# Spearman Rank Correlation ( $\rho$ ) between FSAR Metrics and AGB



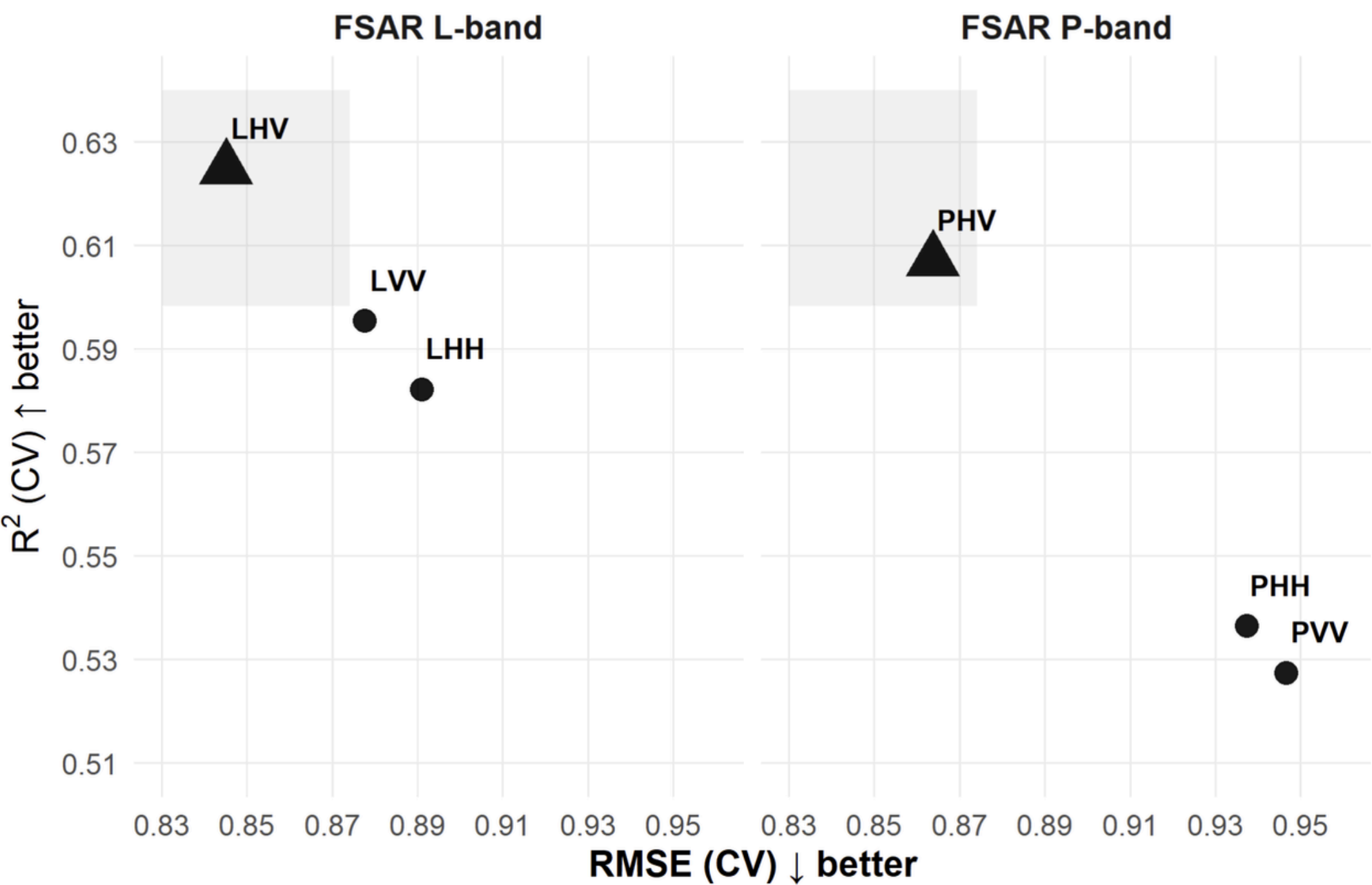
Moderate correlations indicate that radar backscatter is related to forest structure but is also influenced by other factors such as moisture, canopy density and signal saturation.

Moderate correlation with AGB ( $\rho \approx 0.56-0.62$ )  
Strong intercorrelation among HH and HV ( $\rho > 0.90$ )



# Baseline models (Linear Regression, 5-fold CV)

Higher R<sup>2</sup> and lower RMSE indicate better single-predictor performance



# Linear Regression per Polarization Channel

## WHY SINGLE-PREDICTOR MODELS?

POLARIZATION CHANNELS ARE STRONGLY CORRELATED AND USING THEM TOGETHER CAUSES MULTICOLLINEARITY THEREFORE BASELINE MODELS WERE BUILT USING SINGLE PREDICTORS

|   | model | intercept | RMSE      | Rsquared  | MAE       |
|---|-------|-----------|-----------|-----------|-----------|
| 1 | lvv   | TRUE      | 0.8775176 | 0.5954574 | 0.6971430 |
| 2 | lhh   | TRUE      | 0.8909413 | 0.5822906 | 0.7051575 |
| 3 | lhv   | TRUE      | 0.8450224 | 0.6248450 | 0.6618201 |
| 4 | pvv   | TRUE      | 0.9465612 | 0.5275289 | 0.7651824 |
| 5 | phh   | TRUE      | 0.9372842 | 0.5365503 | 0.7355061 |
| 6 | phv   | TRUE      | 0.8636245 | 0.6071025 | 0.6837873 |

BOTH BANDS SHOW COMPARABLE PERFORMANCE.

# Random Forest Regression

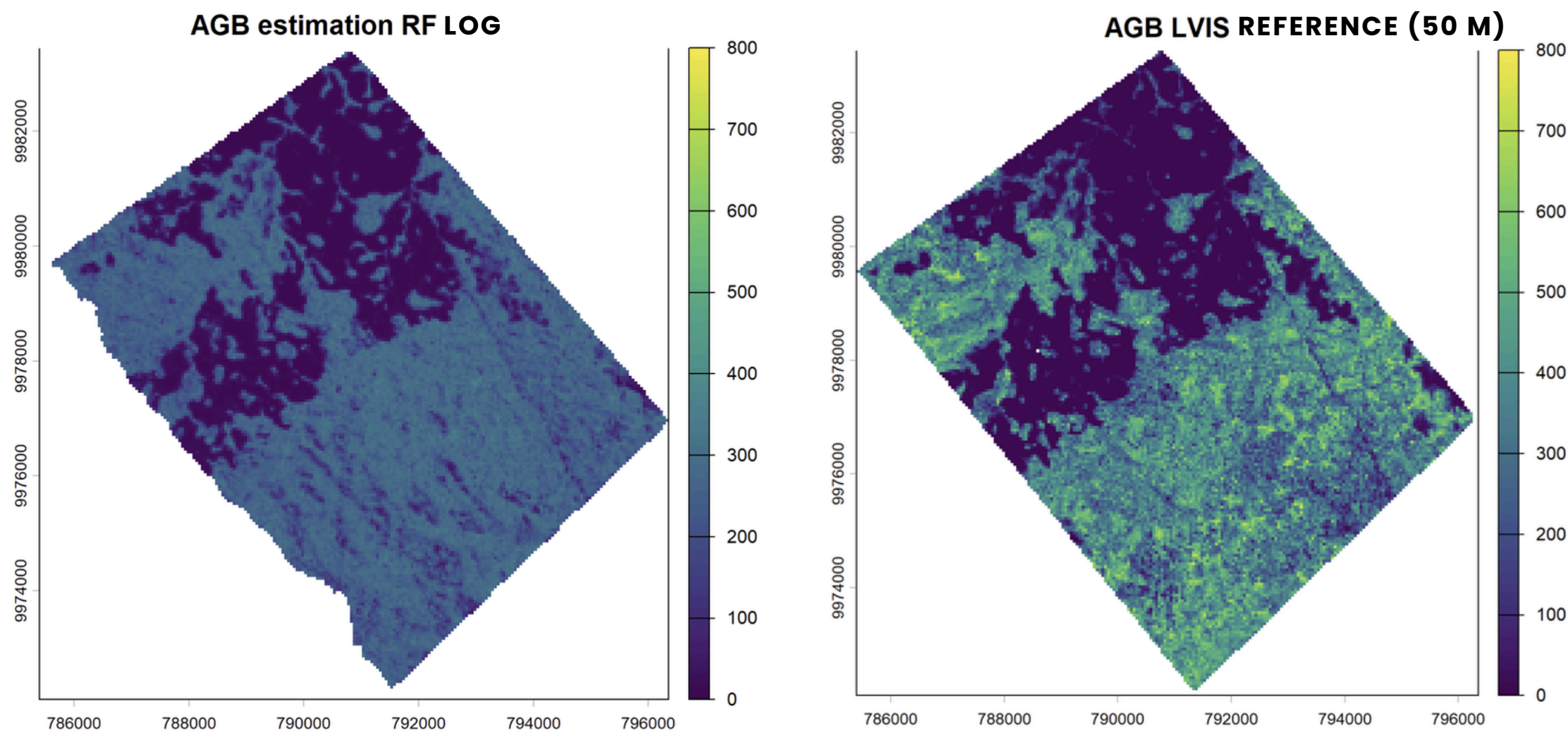
BEST PERFORMANCE ACHIEVED FOR MTRY = 2 (LOWEST RMSE)

| MTRY | RMSE      | R <sup>2</sup> | MAE       |
|------|-----------|----------------|-----------|
| 2    | 0.7192565 | 0.7259027      | 0.5018220 |
| 4    | 0.7244417 | 0.7221587      | 0.5038944 |
| 6    | 0.7278851 | 0.7196890      | 0.5061625 |

The model explains ~73% of the variability in AGB.  
Lower mtry values performed better, likely due to correlated radar predictors.



# Spatial Comparison of Random Forest AGB Predictions and LVIS Biomass



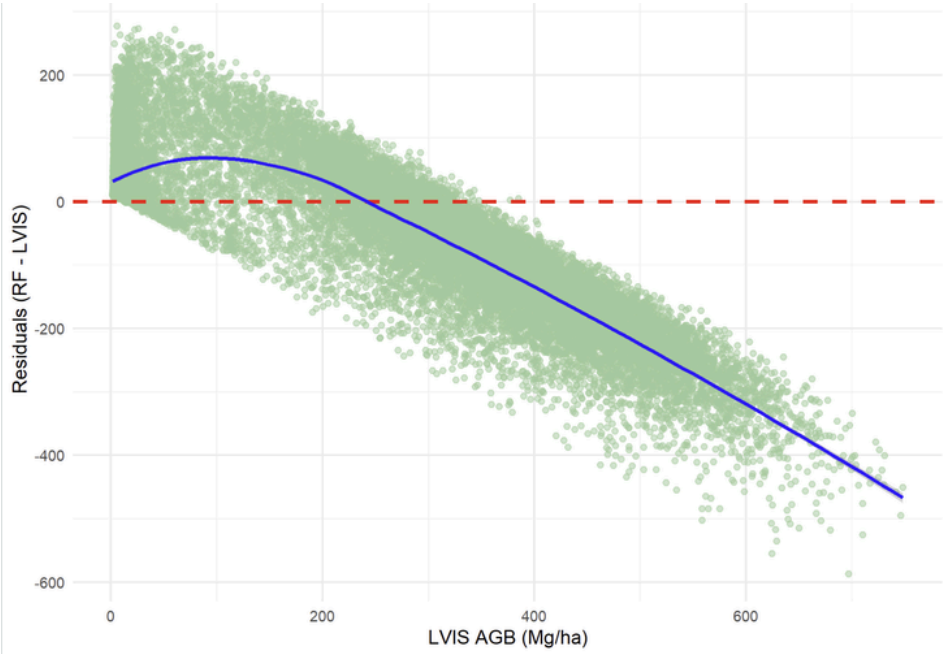
Spatial validation metrics

|          |           |
|----------|-----------|
| N PAIRS  | 23539     |
| RMSE     | 139.0923  |
| RSQUARED | 0.6316548 |
| BIAS     | -55.55531 |
| MAE      | 98.01766  |

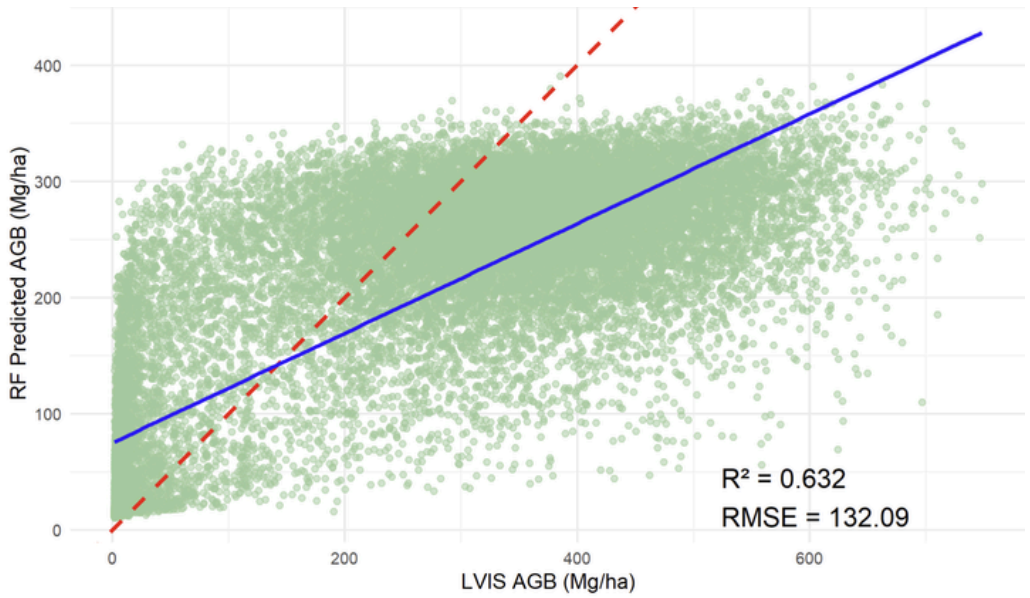
THE RANDOM FOREST MODEL EXPLAINS 63% OF THE SPATIAL VARIABILITY IN AGB ( $R^2 = 0.63$ ).  
RMSE = 139 MG/HA AND MAE = 87 MG/HA INDICATE MODERATE PREDICTION ERRORS TYPICAL FOR SAR-BASED BIOMASS ESTIMATION.  
THE NEGATIVE BIAS (-55 MG/HA) SUGGESTS THAT THE MODEL SLIGHTLY UNDERESTIMATES HIGH BIOMASS AREAS, LIKELY DUE TO RADAR SIGNAL SATURATION IN DENSE FORESTS

What are our results?

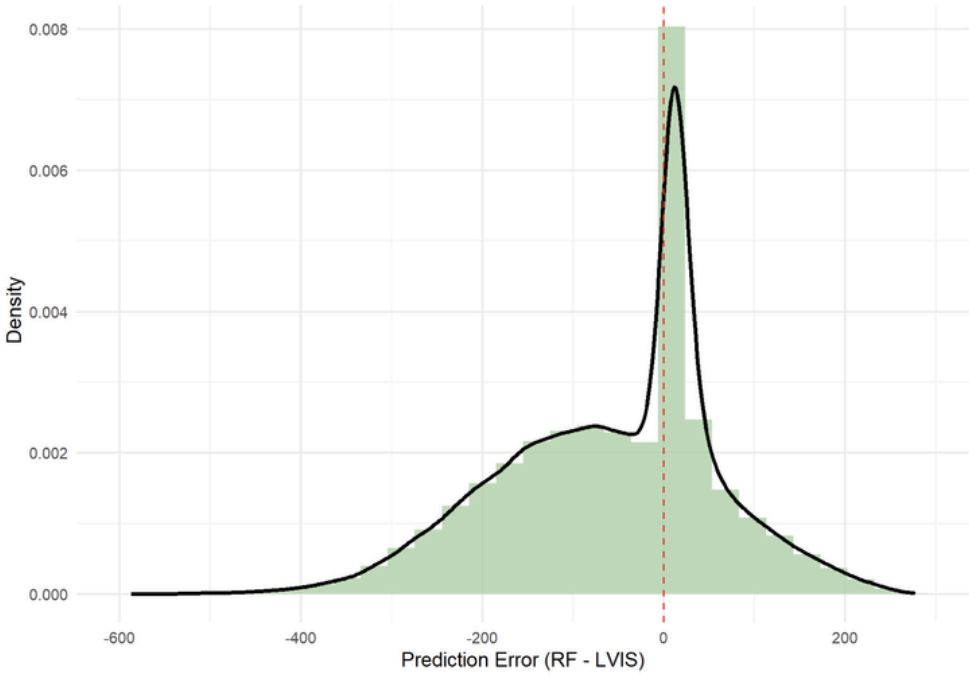
## Residuals vs LVIS AGB



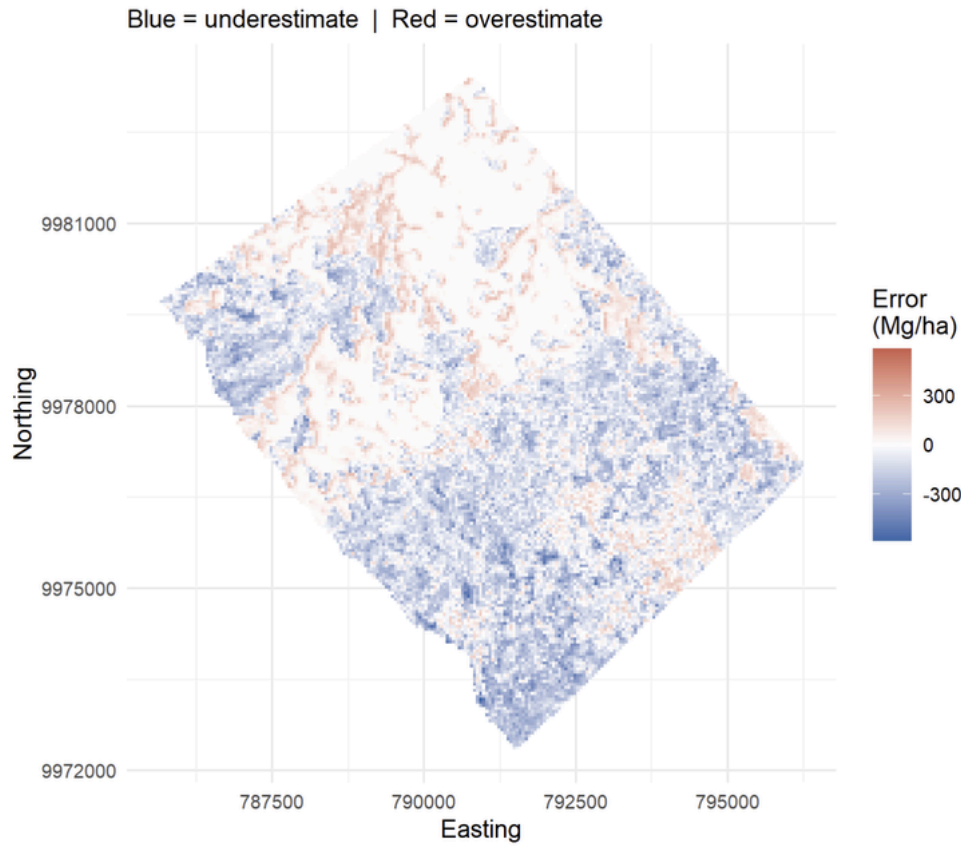
## RF Predictions vs AGB



## Distribution of Prediction Error



## Spatial distribution of Prediction Error



# Random Forest Performance Evaluation

THE RESIDUAL ANALYSIS INDICATES A SYSTEMATIC BIAS WHERE THE MODEL TENDS TO UNDERESTIMATE HIGHER BIOMASS VALUES.

THE ERROR DISTRIBUTION IS CENTERED SLIGHTLY BELOW ZERO, CONFIRMING THIS NEGATIVE BIAS.

THE PREDICTED BIOMASS MAP CAPTURES THE MAIN LANDSCAPE PATTERNS SEEN IN THE LVIS DATA.

**OVERALL, THE MODEL REPRODUCES GENERAL BIOMASS VARIABILITY BUT STILL STRUGGLES WITH HIGH-BIOMASS AREAS, SUGGESTING ROOM FOR IMPROVEMENT IN MODEL CALIBRATION OR PREDICTOR VARIABLES.**

## Precision of Results?



# Final takeaways

1. The Random Forest model captures the general spatial pattern of AGB but shows moderate error
2. High biomass areas remain challenging and tend to be underestimated

# Future improvements

1. Reducing underestimation of the model using more and better training data
2. Trying other prediction models
3. Uncertainty Analysis for gedi data



RESULTS

**Thank you!**





# References

1. Ma, T., Zhang, C., Ji, L., Zuo, Z., Beckline, M., Hu, Y., Li, X., & Xiao, X. (2024). Development of forest aboveground biomass estimation, its problems and future solutions: A review. *Ecological Indicators*, \*159\*, 111653.  
<https://doi.org/10.1016/j.ecolind.2024.111653>
2. Tian, L.; Wu, X.; Tao, Y.; Li, M.; Qian, C.; Liao, L.; Fu, W. Review of Remote Sensing-Based Methods for Forest Aboveground Biomass Estimation: Progress, Challenges, and Prospects. *Forests* 2023, 14, 1086. <https://doi.org/10.3390/f14061086>
3. Flores, A., Herndon, K., Thapa, R., & Cherrington, E. (2019). Synthetic Aperture Radar (SAR) Handbook: Comprehensive Methodologies for Forest Monitoring and Biomass Estimation. <https://doi.org/10.25966/NR2C-S697>
4. Liu, X., Neigh, C. S. R., Pardini, M., & Forkel, M. (2024). Estimating forest height and above-ground biomass in tropical forests using P-band TomoSAR and GEDI observations. *International Journal of Remote Sensing*, 45(9), 3129–3148.  
<https://doi.org/10.1080/01431161.2024.2343134>
5. Kim, Y., & van Zyl, J. J. (2009). A Time-Series Approach to Estimate Soil Moisture Using Polarimetric Radar Data. [DOI:10.1109/TGRS.2009.2014944](https://doi.org/10.1109/TGRS.2009.2014944)

